Topic 3 — Drainage System

Covers syllabus (click to expand)

Himalayan Rivers: Indus (सिंधु) System | Ganga (गंगा) System | Brahmaputra (ब्रह्मपुत्र) System | Trans-Himalayan Rivers | River Capture (River Piracy) | Panch Prayag (प्रयाग) | Rivers of Uttarakhand (उत्तराखंड) | Tributaries **Peninsular Rivers:** Godavari (गोदावरी) | Krishna (कृष्णा) | Kaveri (कावेरी) | Mahanadi (महानदी) | Narmada (नर्मदा) | Tapi (तापी) | Luni (लूनी) | Inland Rivers | Inland Drainage | Fault Valley Drainage **Uttar Pradesh (उत्तर प्रदेश):** Rivers of Uttar Pradesh **River Features & Patterns:** Deltas | Types of Deltas (Bird's Foot, Arcuate (चापाकार), Cuspate, Moribund) | Estuaries | Estuaries vs Deltas | River Origin | Confluence | Drainage Patterns | River Rejuvenation | Structural (संरचनात्मक) Basins **Waterways & Linking:** River Linking Project | River Interlinking | National Waterways | Inland Waterways (NW-1 to NW-5 and New Waterways)

Sources baked in: NCERT Geography Class 11 (Ch 3), Class 12 (Ch 2–3), drainage map (Kullar), UPPCS Prelims PYQs 2018–2025 **Weight:** ★★★ — Prayag/confluence, tributary traps, west-flowing rivers, deltas, UP rivers, dam–river **Last verified:** August 2026 **Current Affairs:** Ken–Betwa = **only NPP link under implementation** (foundation Dec 2024; ~₹44,605 cr; Bundelkhand MP–UP; Panna TR)

Current Affairs (this topic)

Year	Fact	Why asked	Source
Dec 2021 / Dec 2024	Ken–Betwa (बेतवा) Link: Cabinet ₹44,605 cr; PM foundation **25 Dec 2024**	First ILR under implementation	PIB
2025–26	**Only** priority NPP link in implementation (30 identified: 16 peninsular + 14 Himalayan); NWDA nodal	"How many links operational?" trap	Rajya Sabha (राज्यसभा) / Jal Shakti (शक्ति) Aug 2026
2026	Spend ~₹12,382 cr to Jun 2026; target ~Mar 2030; Bundelkhand MP–UP; **Panna TR** forest	Cost / ecology (पारिस्थितिकी) pairing	LS Q Aug 2026
Ongoing	Polavaram includes Godavari–Krishna (Vijayawada) link assistance	Peninsular grid	PIB
2016	National Waterways Act — 106 NWs; core still NW-1 to NW-5	Number vs names	Act

Namami Gange (नमामि गंगे) and GAP are Ganga cleaning programmes. They are not river-origin facts.

🗺️ Visual Cheat Sheet — Complete Indian Drainage System

INDIAN PHYSICAL GEOGRAPHY – DRAINAGE SYSTEM

COMPLETE SOURCE OF TRUTH – UPPCS/UPSC PRELIMS & MAINS

1 QUICK REVISION – RAATA THIS FIRST

India Drainage – 4 Major Himalayan Systems + Peninsular Rivers

- Indus – 3180 km (mostly in Pakistan)
- Ganga – 2525 km; basin ~8.6 lakh km² (largest in India)
- Brahmaputra – 2900 km (Yarlung Tsangpo in Tibet)
- Peninsular Rivers – old, graded, rain-fed

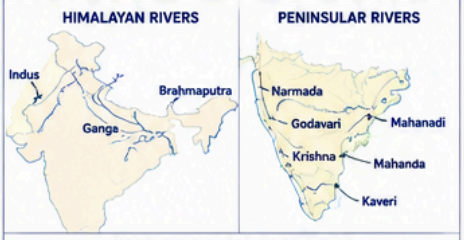
Key Contrasts

HIMALAYAN RIVERS	PENINSULAR RIVERS
Youth, vigorous	Mature, graded
Perennial	Mostly seasonal
V-shaped valleys	Shallow valleys
Meanders, floodplains	Fixed course
Large deltas	Estuaries (west flowing)

IMD Seasons (Hydrology view)

- Cold Weather: Dec – Feb
- Hot Weather: Mar – May
- SW Monsoon / Rainy: Jun – Sep
- Retreating: Oct – Nov
- NE Monsoon (SE coast only): Oct – Dec

2 MAJOR DRAINAGE SYSTEMS OF INDIA



BASIN AREA & LENGTH (High Yield)

River System	Basin Area (approx.)	Length (approx.)
Ganga	~8.6 lakh km² (Largest in India)	2525 km
Godavari	~3.1 lakh km² (Largest Peninsular)	1465 km (Longest Peninsular)
Brahmaputra	~5.8 lakh km²	2900 km
Indus	~3.2 lakh km² (in India part)	3180 km
Krishna	~2.6 lakh km²	1400 km
Mahanadi	~1.4 lakh km²	851 km
Narmada	~0.98 lakh km²	1312 km (Longest West-flowing)
Tapi	~0.65 lakh km²	724 km
Kaveri	~0.81 lakh km²	800 km

TRANS-HIMALAYAN RIVERS: Indus, Sutlej, Brahmaputra (Tsangpo), Zanskar
MCQ = Sutlej (2020 Q52)

3 PANCH PRAYAG (HIMALAYAN CONFLUENCES)

Order (Upstream to Downstream)	Confluence	Rivers Meet	On	Forms
1	Vishnu Prayag	Dhauli Ganga + Alaknanda	Joshimath	
2	Nanda Prayag	Nandakini + Alaknanda	Nandaprayag	
3	Karn Prayag	Pindar + Alaknanda	Karnaprayag	
4	Rudra Prayag	Mandakini + Alaknanda	Rudraprayag	
5	Dev Prayag	Bhagirathi + Alaknanda	Devprayag	GANGA

Memory Trick: V → N → K → R → D
Vishnu → Nanda → Karn → Rudra → Dev
(Dev = Done → Ganga is born) [2023 Q50, 2024 Q74]

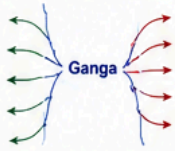
4 GANGA SYSTEM – TRIBUTARIES

LEFT BANK TRIBUTARIES (Stand facing downstream)

- Ramganga
- Gomti
- Ghaghra
- Gandak
- Kosi
- Mahananda

RIGHT BANK TRIBUTARIES (Stand facing downstream)

- Yamuna
- Son
- Damodar
- Punpun
- Jalangi



Jonk = Mahanadi tributary NOT Ganga (2021 Q5)

5 YAMUNA TRIBUTARIES (West → East)



- Order (2020 Q68) = 1, 2, 3, 4 ✓
- Betwa joins Yamuna at Hamirpur (UP) – NOT Prayagraj (2023 Q55)
- Ramganga joins Ganga at Kannauj (UP)

6 ORIGIN OF MAJOR PENINSULAR RIVERS (2023 Q53 = A)

River	Origin
Yamuna	Yamunotri (Uttarakhand)
Krishna	Mahabaleshwar (Maharashtra)
Godavari	Trimbakeshwar, Nashik (Maharashtra)
Mahanadi	Sihawa Hills (Chhattisgarh)
Narmada	Amarkantak (MP)
Tapi (Tapi/Tapti)	Multai (Satpura range, MP) NOT Western Ghats (2018 Q102)
Kaveri	Brahmagiri Hills (Karnataka)
Son	Amarkantak (MP) – radial with Narmada

7 PENINSULAR DRAINAGE – OUTFALL

EAST FLOWING (Bay of Bengal)

Godavari, Krishna, Mahanadi, Kaveri, Pennar, Palar, Damodar (Subarnarekha), Vamsadhara

WEST FLOWING (Arabian Sea)

Narmada, Tapi, Mahi, Periyar (Kerala)



Memory:** Periyar is the only major west flowing river in Kerala (2025 Q116)

8 IMPORTANT TRAPS (High Yield)

- Pranhita → joins Godavari (NOT Mahanadi) (2023 Q49)
- Hemavati → Kaveri tributary
- Srisailam (Krishna) – dam named as per river it is on
- Luni → Rann of Kutch (inland, saline lower course) (2019 Q84)
- Ghaggar → disappears in Thar Desert (inland)
- Barak River → flows to sea (Bay of Bengal), NOT inland (2021 Q65)

9 RIVER FEATURES & PATTERNS

DELTAS (Types)

- Arcuate (Fan shaped) – Sundarbans (Ganga-Brahmaputra)
- Bird's foot (Digitate) – Mississippi (USA)
- Cuspate – Mahanadi Delta
- Estuarine – Narmada, Tapi, Mahi, Periyar
- Moribund (Dying) – Cauvery Delta (2019 Q85)

DRAINAGE PATTERNS

- Dendritic – on uniform rocks (example: Most Peninsular rivers)
- Trellis – in folded/striped rocks (2024 Q31)
- Radial – streams diverge from a central highland (example: Amarkantak)
- Centripetal – streams flow towards a central basin (example: Imphal basin)

10 RIVER CAPTURE, INLAND & REJUVENATION

RIVER CAPTURE (PIRACY)

✓ Teesta captured by Brahmaputra. A/R both true (2021 Q63)



INLAND DRAINAGE

- Luni → Rann of Kutch (saline)
- Ghaggar → Thar Desert depression

RIVER REJUVENATION

Renewed erosive power after uplift or fall in sea level → forms terraces, gorges, waterfalls.

Gomti in UP = Biological Disaster! (2018 Q99)

11 WATERWAYS, LINKING & MAJOR DAMS

NATIONAL WATERWAYS (NW)

- NW-1 : Ganga – Hooghly (Allahabad to Haldia)
- NW-2 : Brahmaputra (Dhubri to Sadiya)
- NW-3 : West Coast Canal, Kerala (NW-3 Kollam to Kottapuram)
- NW-4 : Krishna – Godavari Canal (Dellang to Puducherry)
- NW-5 : Brahmani – Mahanadi (Talcher to Dhamra)

NEW MULTI-MODAL WATERWAYS

- NW-6 : Barak (Assam)
- NW-7 : Sundarbans Waterways (WB)
- NW-10 : Amba – Arpa – Vainganga, MP

IMPORTANT DAMS (River wise)

Dam	River
Baglihar Dam	– Chenab
Pandoh Dam	– Beas
Srisailam Dam	– Krishna
Hirakud Dam	– Mahanadi
Mettur (Stanley) Dam	– Kaveri
Tehri Dam	– Bhagirathi
Rihand Dam	– Rihand (Son Basin)

RAMGANGA PROJECT (UP) (2024 Q72 = D)

- 198 MW (Garhwal)
- Irrigated area ~5.75 – 5.90 lakh ha (NOT 6 lakh)

★ RAPID FACTS (RAATA)

- ★ Largest basin in India – Ganga
- ★ Largest peninsular basin – Godavari
- ★ Longest peninsular river – Godavari (~1465 km)
- ★ Longest west flowing – Narmada (~1312 km)
- ★ Only river in Kerala flowing to Arabian Sea – Periyar
- ★ Only trellis pattern – in folded mountains
- ★ Radial pattern centre – Amarkantak
- ★ Only moribund delta – Cauvery
- ★ Teesta – captured by Brahmaputra
- ★ Jonk – Mahanadi tributary, NOT Ganga
- ★ Sutlej – Trans-Himalayan (MCQ trap)
- ★ Rann of Kutch – formed by Luni

12 MVET-KNOW TERM COMPARISONS (One-line Difference)

Term	One-line Difference	Hindi
Drainage basin	Land area drained by a river and its tributaries	जल विभाजक क्षेत्र
Watershed / Water divide	Ridge separating two drainage basins	जल विभाजक रेखा
Perennial river	Flows throughout the year (snow-fed + rain-fed)	बारहमासी नदी
Non-perennial / Ephemeral	Flows only after rain; dry channel in dry season	अल्पवार्षिक नदी

Term	One-line Difference	Hindi
River capture	Stronger stream steals headwaters of weaker stream (piracy)	नदी अपहरण
Inland drainage	River does not reach sea; ends in lake/plain/desert	अंतःस्थ जल निकास
Antecedent river	Maintains course across rising land (Indus, Brahmaputra)	पूर्वगत जल निकास
Superimposed river	Maintains course after burying relief (Chambal)	अध्यारोपित नदी

Term	One-line Difference	Hindi
Fault valley river	Course in tectonic rift/graben (Narmada, Tapi, Damodar)	भ्रंश घाटी नदी
Estuary	Drowned river valley; tide-dominated, sediments removed	ज्वारनदमुख
Delta	Deposited sediment at river mouth; prograded into sea	डेल्टा / त्रिभुज
Moribund delta	Old inactive delta lobe no longer receiving sediment	मृत डेल्टा

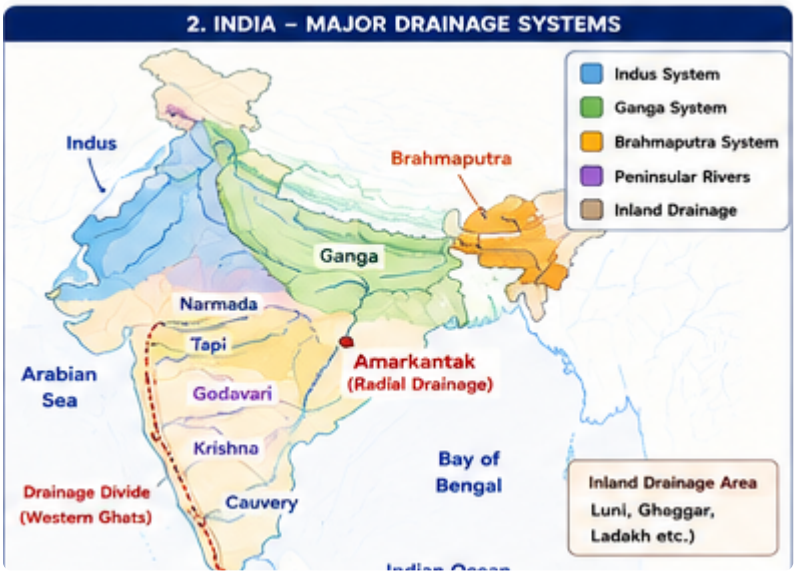
Term	One-line Difference	Hindi
River rejuvenation	Renewed erosive power after uplift / sea-level fall	नदी पुनः युवा अवस्था
Structural basin	Lowland enclosed by hills; centripetal drainage	संरचनात्मक द्रोणी
Prayag	Sacred Himalayan confluence (Uttarakhand)	प्रयाग
Sangam	Confluence of Ganga + Yamuna at Prayagraj	संगम

UPPCS/UPSC EXAM WEIGHT: ★★★ HIGH | DRAINAGE, TRIBUTARIES, PRAYAG, DELTAS, DAM PAIRS – | LAST VERIFIED: AUGUST 2026

★ UPPCS/UPSC EXAM WEIGHT: ★★★ HIGH | REPEATED EVERY YEAR | LAST VERIFIED: AUGUST 2026

3.0 Drainage framework

River basics (must know before systems)



- ~8% → Inland Drainage

⚠ **Main Drainage Divide**
- Western Ghats (Peninsula)
- Amarkantak (Central India)

⚠ **Basin Highlights**
- Ganga – Largest in India
- Godavari – Largest Peninsular
- Mahanadi – Among Tapi, Narmada, Mahanadi, Cauvery → Mahanadi

⚠ **Nature of Rivers**
- Himalayan – Perennial, antecedent
- Peninsular – Rain-fed, well graded

⚠ **Inland Drainage (Endorheic)**
- Luni (Rajasthan), Ghaggar,
- Sambhar, Salt Lake, Ladakh

- A **drainage basin** is the whole catchment of a large river. A **watershed** is a smaller unit inside that basin.
- A **major** basin has a catchment above **20,000 km²**. A **medium** basin is 2,000–20,000 km². A **minor** basin is below 2,000 km².
- About **77% of India's drainage area** faces the **Bay of Bengal**. About **23%** faces the **Arabian Sea**. Inland basins cover about **8% of area** (Luni, Ghaggar (घग्गर), Ladakh (लद्दाख) closed basins).
- **Water volume is more lopsided than area.** Over **90% of river water** goes to the Bay of Bengal. The Brahmaputra alone contributes about **40%** of India's river water. The Ganga contributes about **25%**.

Outlet	Share of area	Typical rivers
Bay of Bengal	about 77%	Ganga, Brahmaputra, Godavari, Krishna, Mahanadi, Kaveri, Pennar, Palar, Vaigai, Damodar, Subarnarekha, Barak
Arabian Sea	about 23%	Indus (in Pakistan), Narmada, Tapi, Mahi, Sabarmati, **Periyar**, Sharavati, Mandovi
Inland	about 8% of area	Luni, Ghaggar, Rajasthan salt (लवणाध्यक्ष) lakes, Ladakh

- **Ganga** is the largest basin **inside India** (about **8.61 lakh km²**, about 26% of India).
- **Godavari** is the largest **peninsular** basin (about **3.13 lakh km²**).
- Among Tapti / Narmada / Mahanadi / Cauvery, the largest basin is **Mahanadi**. Godavari is not in that set.

Himalayan vs peninsular rivers

- The **Himalayan drainage** includes the three great systems: **Indus, Ganga, and Brahmaputra**.
- Himalayan rivers are young and mostly **perennial**. They get water from **glaciers, springs, and monsoon rain**.
- In the mountains they cut **gorges, waterfalls, and steep valleys**. In the plains they build floodplains, meanders, and deltas.
- **Peninsular rivers** are older and **graded**. Most are **rain-fed** and more seasonal. Courses are fixed. Meanders are fewer than in the Himalaya.
- East-flowing peninsular rivers build **deltas**. West-flowing **Narmada and Tapi** (rift valleys) build **estuaries**.

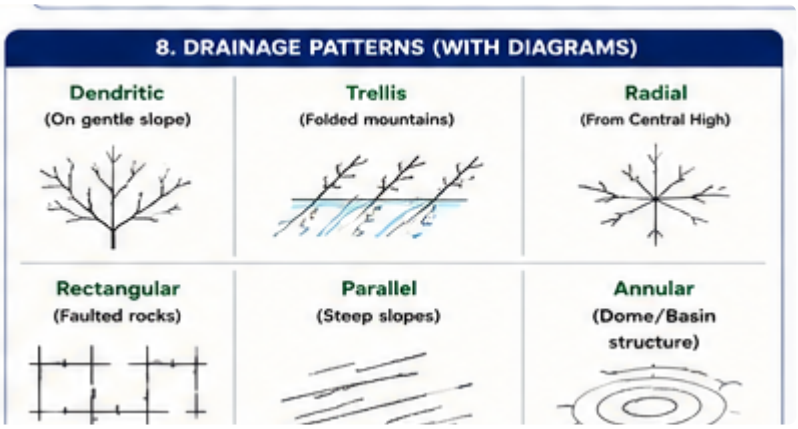
Drainage types (definitions) — Teaching Card

Genetic drainage names describe **what the river follows** or **when it formed relative (सापेक्ष) to uplift**.

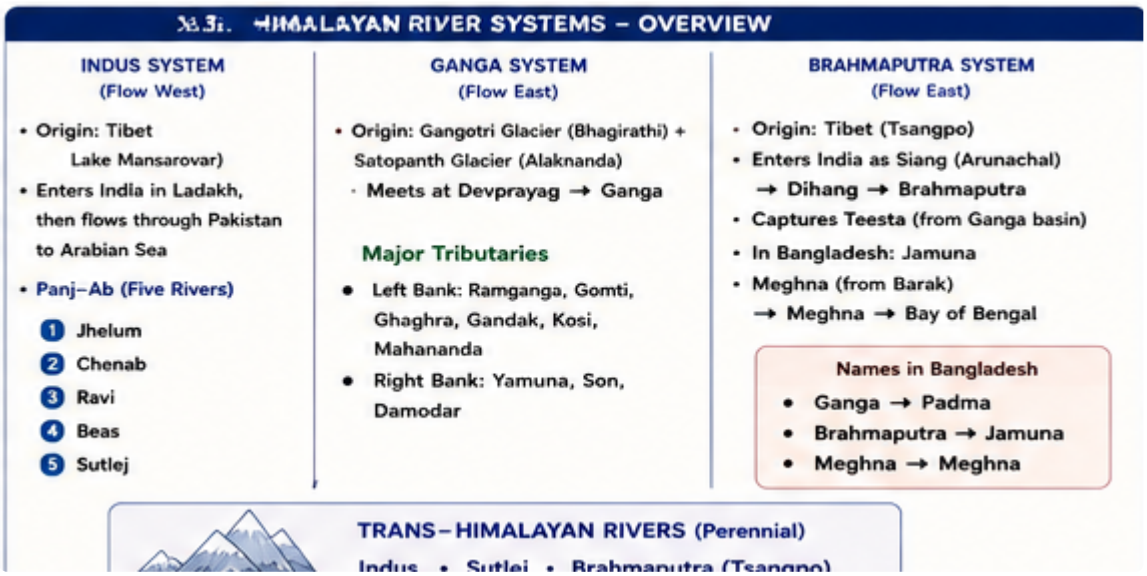
- A **consequent** river follows the **original slope** of the land — Godavari, Krishna and Kaveri draining from the Western Ghats are Indian examples.
- A **subsequent** river grows later along a **weak belt** and often joins at right angles — Chambal, Sind, Betwa, Ken, Tons (टौंस) and Son toward Yamuna / Ganga.
- An **antecedent** river is **older than uplift**; it keeps its course by cutting a gorge through rising land — classic set **Indus, Sutlej, Brahmaputra**; many keys also tag **Ganga**.
- A **superimposed** river inherits an older course onto newer structure and does **not** follow today's slope — **Chambal** (चंबल) (also Son in some notes).
- **Chambal can carry both labels. Subsequent** answers stems about a weak-belt tributary joining the Yamuna at right angles. **Superimposed** answers stems about an inherited course that does not follow present structure. Match the **word in the stem**; do not treat the two tags as mutually exclusive facts about different rivers.
- **Antecedent** is genetic history (इतिहास) (river older than Himalayan uplift). **Dendritic** is plan-view pattern on homogeneous rock or alluvium. The Ganga can be **antecedent** in origin keys and still show **dendritic** drainage on the middle–lower plain — those are different questions, not a contradiction.

- **Concordant** drainage follows structure and slope — most peninsular rivers.
- **Discordant** drainage cuts **across** structure — Himalayan antecedent rivers.
- A **river capture (piracy)** happens when a stronger stream steals the headwaters of a weaker one. **Teesta** shifting from Ganga to Brahmaputra is the UPPCS fact.

Type	Meaning	India
Consequent	Follows the original slope of the land	Godavari, Krishna, Kaveri from Western Ghats
Subsequent	Grows later along a weak belt; often joins at right angles	Chambal, Sind, Betwa, Ken, Tons, Son to Yamuna/Ganga
Antecedent	Older than uplift; keeps course by cutting a gorge through rising land	Classic set: **Indus, Sutlej, Brahmaputra**; many PCS keys also tag **Ganga**
Superimposed	Older course inherited onto newer structure; does not follow present slope	**Chambal** (also Son in some notes)
Concordant	Follows structure and slope	Most peninsular rivers
Discordant	Cuts across structure	Himalayan antecedent rivers



Trans-Himalayan rivers (separate tag)

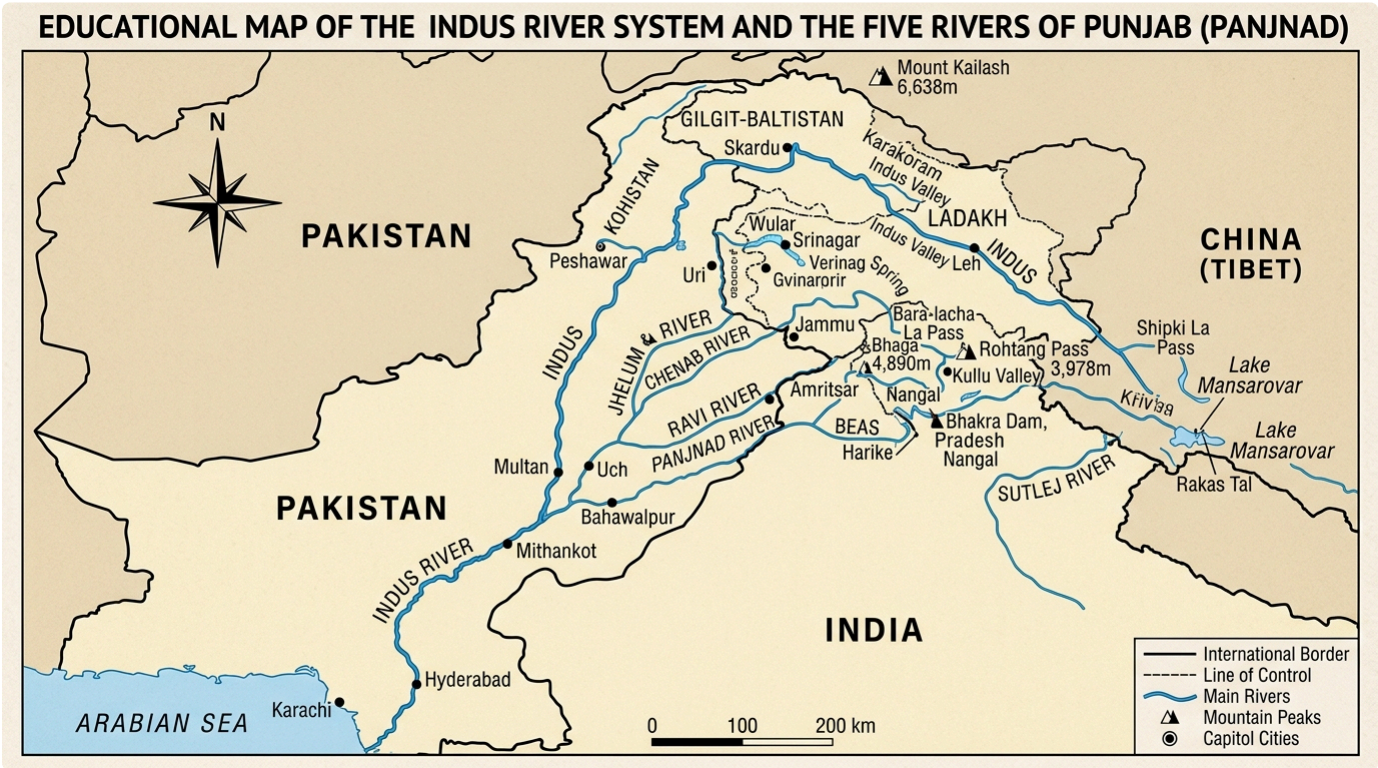


- **Trans-Himalayan rivers** rise **north of the Great Himalaya**, usually on the **Tibetan Plateau**, then **cut through** the Himalaya into India.
- They are typically **antecedent**. Origin is beyond the main Himalayan crest, not in the Himachal (हिमाचल) / Kashmir Himalaya proper.
- set: **Indus, Sutlej, and Brahmaputra (Tsangpo)**.
- Among common options (Jhelum / Sutlej / Ganga / Ravi), the Trans-Himalayan pick is **Sutlej**.
- **Sutlej** rises at **Rakas Tal** near Mansarovar (Tibet), enters India near **Shipki La** (शिपकी ला), and gorges through the Himalaya. That origin-plus-cut-through is why papers ask "Sutlej is Trans-Himalayan."
- **Jhelum, Ravi, Beas, and Ganga** (formed at Devprayag) are **not** the Trans-Himalayan pick in that MCQ set.
- The peninsula's water divide is the **Western Ghats**. **Amarkantak** is a **radial** node: Narmada goes west, Son goes to the Ganga.

> **Logic:** Among Tapi / Narmada / Mahanadi / Cauvery, the largest basin is **Mahanadi**. Himalayan snow feeds perennial rivers — assertion and reason both hold, and the reason explains the assertion. Trans-Himalayan is **not** every Himalayan river.

3.1 Himalayan rivers

Indus system



- The Indus rises near the **Kailash / Mansarovar** region of Tibet (Bokhar Chu class origin). It is **antecedent** and **Trans-Himalayan**. It cuts a gorge in Ladakh and reaches the **Arabian Sea** near

Karachi.

- Total length is about **2900–3180 km**. Only about **710 km** lies in India.
- **Shyok** and **Zanskar** (ज़ांस्कर) join in Ladakh. **Gilgit** joins farther west.
- The five eastern tributaries are the **Panj Ab**. From **east to west** they are **Sutlej, Beas, Ravi, Chenab, Jhelum**.
- They meet as **Panjnad** just above **Mithankot**, then join the Indus.
- The **Jhelum** (झेलम) rises at **Verinag**. It flows through **Wular Lake** and joins the Chenab. **Kishanganga** (Neelum in Pakistan) is a Jhelum tributary.
- The **Chenab** forms near **Bara Lacha La** (Chandra + Bhaga (भाग) in Lahaul). **Baglihar** and **Salal** stand on the Chenab.
- The **Ravi** rises in the Kullu hills near **Rohtang** (रोहतांग). It drains between Pir (पीर) Panjal (पीर पंजाल) and Dhauladhar (धौलाधार). **Ranjit Sagar** is on the Ravi. **Pandoh is not** on the Ravi.
- The **Beas** rises near **Rohtang**. **Pandoh Dam** is on the Beas. Beas meets Sutlej near **Harike**.
- The **Sutlej** rises at **Rakas Tal**, Tibet. It is both **Trans-Himalayan** and **antecedent**. It enters Himachal near **Shipki La** and cuts deep gorges across the Himalayan ranges. **Bhakra–Nangal** stands on the Sutlej. Among Jhelum / Sutlej / Ganga / Ravi, the Trans-Himalayan answer is **Sutlej**.
- Join order toward the Indus: the **Jhelum** and **Ravi** join the **Chenab**. The Chenab then meets the **Sutlej**. The Sutlej finally joins the **Indus** (सिंधु).
- **Indus Waters Treaty (1960):** India uses **Ravi, Beas, Sutlej**. Pakistan uses **Indus, Jhelum, Chenab**.

> **Logic: Pandoh–Ravi** and **Srisailam–Tungabhadra** are the classic NOT-matched pairs. Among usual Himalayan options, the Trans-Himalayan river is the **Sutlej**.

PYQ — UPPCS Prelims 2025, Q92

Which of the following pairs is/are NOT correctly matched? (Dam) — (River)

1. Baglihar Dam — Chenab
2. Pandoh Dam — Ravi
3. Srisailam Dam — Tungabhadra

A. 1 and 2

B. Only 3

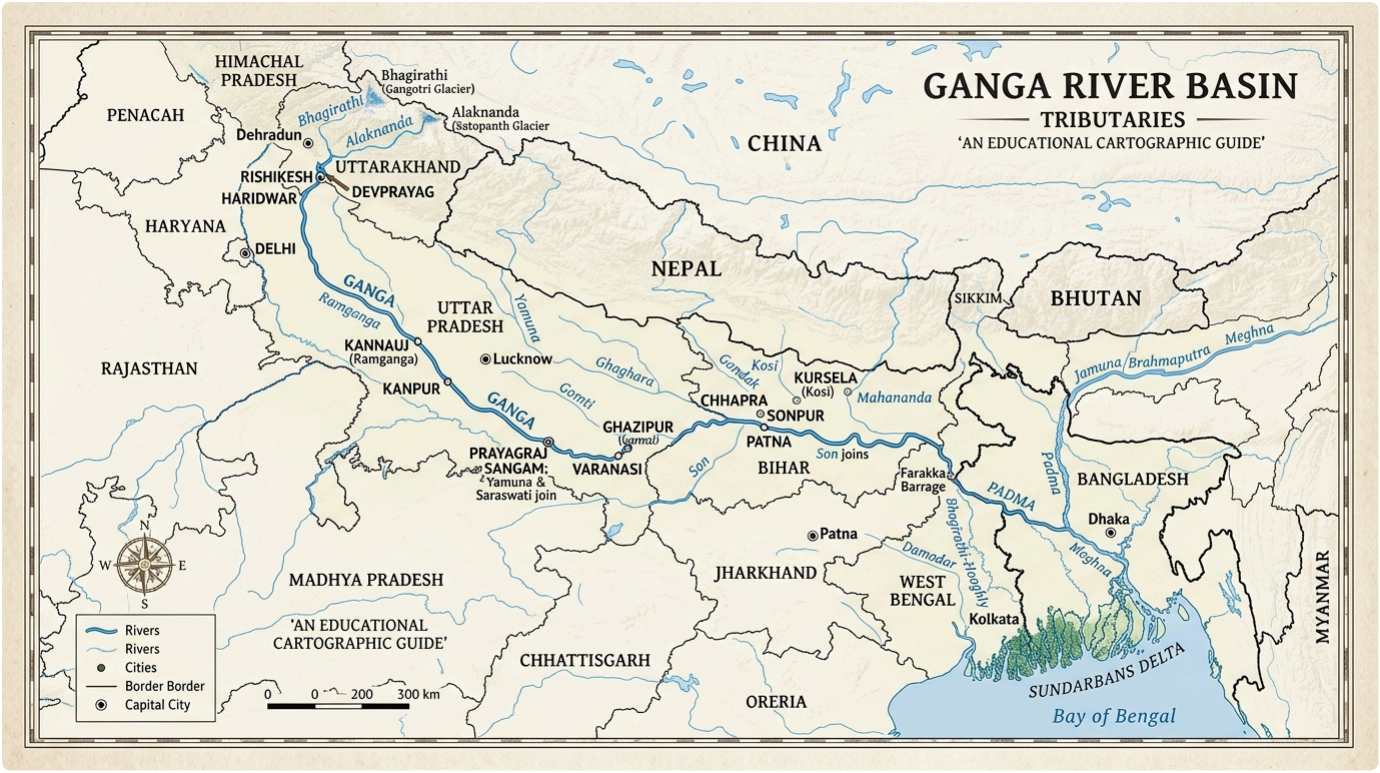
C. 2 and 3

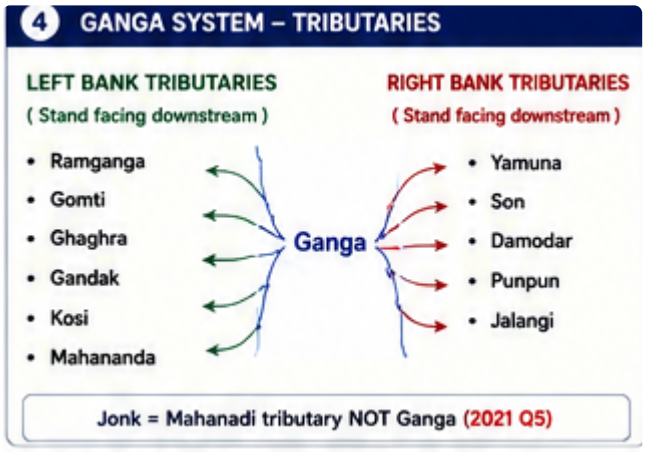
D. Only 1

Show answer

Ans: C — Pandoh = **Beas**; Srisailam = **Krishna**. Baglihar–Chenab is correct.

Ganga system





- The Ganga is formed at **Devprayag** where **Alaknanda** meets **Bhagirathi** (भागीरथी). Length is about **2525 km** — India's longest river.
- The **Bhagirathi** rises at **Gomukh** (Gangotri (गंगोत्री) glacier). The **Alaknanda** rises near **Satopanth** above Badrinath. **Badrinath** sits on the Alaknanda. **Kedarnath** (केदारनाथ) sits on the Mandakini.
- The Ganga enters the plains at **Haridwar** (हरिद्वार). At **Farakka** water is diverted into the **Hooghly** (Bhagirathi–Hooghly arm in West Bengal). The main stem enters Bangladesh near Farakka as the **Padma**.
- In Bangladesh the Padma is joined by the **Jamuna** (Brahmaputra). Further downstream the Padma meets the **Meghna**. From that join onward the combined outlet is commonly called the **Meghna system**, which drains into the **Bay of Bengal**.
- The world's largest delta is the **Ganga–Brahmaputra–Meghna (Sundarbans)** delta. It lies between the **Hooghly** in the west and the **Meghna** in the east.
- The Ganga is often cited as an **antecedent** river. The classic antecedent set remains **Indus, Sutlej, Brahmaputra**; many PCS keys also tag Ganga.

- **Left-bank (Himalayan) tributaries** join from the north: Ramganga, Gomti, Ghaghara, Gandak, Kosi, Mahananda.
- **Right-bank tributaries** come mainly from the peninsula: Yamuna, Son, Damodar (via Hooghly).

In-depth Tributary Profiles & Ganga Traps

- **Ganga Alluvial Depth:** According to R.D. Oldham, the depth of alluvium in the Indo-Gangetic trough is about **4,000 to 6,000 metres**, while gravity surveys by E.A. Glennie estimated it at ~**2,000 metres**.
- **Son River Origin & Drainage:** Rises on the Amarkantak plateau (Sonmuda in Anuppur district, MP; geologically also recognized from Sonbachharwar/Sonekund in Pendra Road, Gaurela-Pendra-Marwahi, CG). Its catchment area is **70,055 sq km**, highest among major Ganga feeders. It flows northward to meet the Ganga near Patna.
- **Kaimur Range:** Acts as a decisive watershed separating the Son basin on the south from the Tons (Yamuna system) on the north.
- **Gandak (Narayani) & Triveni Canal:** Originates in Nepal Himalayas between Dhaulagiri and Everest as the Kaligandak and Trishulganga. Known as the **Gandaki** or **Narayani** in Nepal. The **Triveni Canal** is constructed on the Gandak in West Champaran (Bihar). It joins the Ganga at Sonpur near Patna and does **not** flow through Uttar Pradesh.
- **Ganga in Bihar:** Traverses **12 districts** (Saran, Buxar, Bhojpur, Patna, Vaishali, Samastipur, Begusarai, Lakhisarai, Munger, Khagaria, Bhagalpur, Katihar). Length is longest in **Patna (~99 km)** and second in **Bhagalpur (~97 km)**.
- **Punpun River:** Originates in Chotanagpur Plateau (Jharkhand), flows through Aurangabad district of Bihar, and merges into the Ganga at **Fatuha** (Patna district).
- **Kosi Plain:** Formed by the confluence of 7 streams (principal: Arun Kosi, Sun Kosi, Tamur Kosi). Drains northern slopes of Himalayas in Tibet and southern slopes in Nepal; does **not** have its origin in glaciers. Joins Ganga at Kursela (Katihar). Average elevation of Kosi plain is **30 to 40 metres**.
- **Kanwar Lake (Begusarai):** Asia's largest freshwater oxbow lake, formed by a cutoff meander of the **Burhi Gandak** River, NOT the main Gandak.

Where major rivers meet the Ganga

Tributary	Bank	Meets Ganga at / near
Yamuna (यमुना)	Right	**Prayagraj (Sangam)**
Ramganga (रामगंगा)	Left	**Kannauj**
Gomti	Left	Near **Saidpur / Ghazipur** belt (after Lucknow)
Ghaghara	Left	**Chhapra / Revilganj (Saran)**
Gandak	Left	**Sonpur near Patna**
Son	Right	Near **Patna**
Punpun	Right	**Fatuha** (Patna)
Kosi	Left	**Kursela, Katihar**
Mahananda (महानंदा)	Left	West Bengal (last left-bank in India)

Tributary	Bank	Meets Ganga at / near
Damodar	—	Joins **Hooghly**, not the main Ganga stem

- The **Ramganga** rises in Garhwal (गढ़वाल) / Doodhatoli. It meets the Ganga at **Kannauj**.
- The **Gomti** rises at **Gomat Taal** (Pilibhit). It is the usual key fact for a Ganga tributary that **rises in the plains**. **Lucknow** (लखनऊ) sits on it. UPPCS 2018 tagged it a **biological disaster**.
- The **Ghaghara / Saryu** (Karnali in Nepal; Mapchachungo glacier class origin) is joined by the **Sharda (शारदा) / Kali**. It meets the Ganga near **Chhapra / Revilganj (Saran)**. **Ayodhya** (अयोध्या) sits on it.
- The **Gandak** (also **Narayani** in Nepal; Kaligandak–Trishuli) joins the Ganga at **Sonpur near Patna**. It does **not** flow through Uttar Pradesh.
- The **Kosi** is the **Sorrow of Bihar**. Sun Kosi, Arun, and Tamur meet in Nepal. It shifts course and floods. It joins the Ganga near **Kursela (Katihar)**.
- The **Mahananda** is the last left-bank tributary in India (West Bengal).
- The **Son** rises at **Amarkantak** (near Narmada's source). It is the main **north-flowing** right-bank Ganga (दायाँ तट) tributary and joins near **Patna**. Rihand (रिहंद) and North Koel are Son tributaries. **Kaimur Range** divides the Son (south) from the Tons / Yamuna system (north).
- The **Damodar** drains Chotanagpur (छोटानागपुर) in a **rift**. It is the **Sorrow of Bengal**. **Barakar** is its main tributary. It meets the Hooghly. Eden Canal is taken from Damodar.
- The **Punpun** joins the Ganga at **Fatuha** (Patna).
- **Jonk** is **not** a Ganga tributary. It joins the **Mahanadi**.

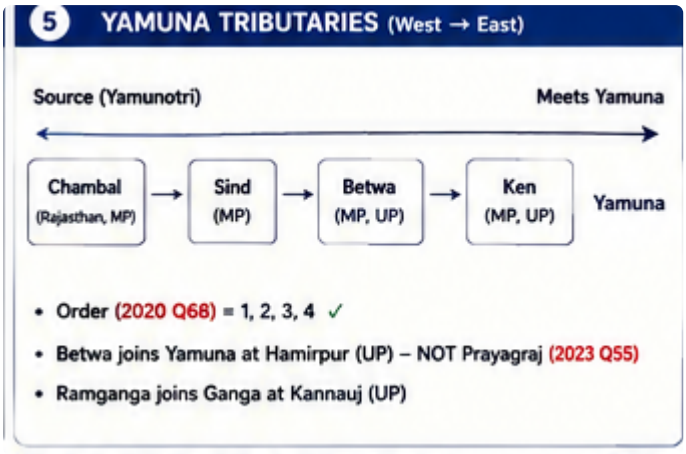
Yamuna system — where tributaries meet

- The **Yamuna** is the longest / largest tributary of the Ganga. It rises at **Yamunotri** on **Bandarpunch** (बंदरपूंछ). It meets the Ganga at **Prayagraj**.

Tributary	Bank of Yamuna	Meets Yamuna at / near
Tons (टौंस)	Left	Near **Kalsi** (Uttarakhand)
Hindon	Left	NCR / western UP belt
Chambal	Right	Near **Etawah**
Sind (सिंध)	Right	Between Chambal and Betwa (Auraiya–Kalpi belt)
Betwa (बेतवा)	Right	**Hamirpur** — **not** Prayagraj
Ken	Right	Near **Banda**

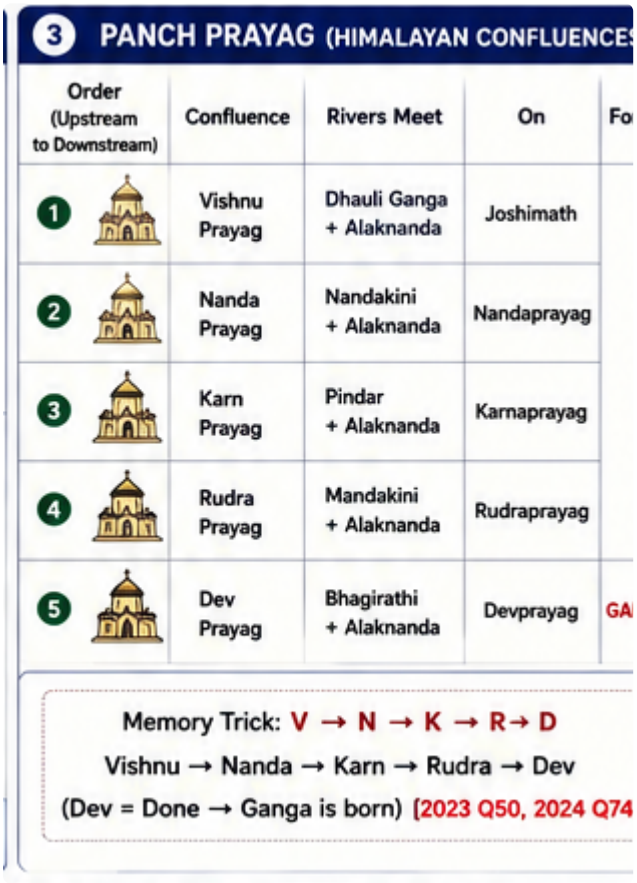
- Yamuna **right-bank** feeders from west to east are **Chambal**, then **Sind**, then **Betwa**, then **Ken**.
- The **Chambal** rises near **Mhow / Dr Ambedkar (अम्बेडकर) Nagar** on the Malwa Plateau (Janapao / Vindhya (विंध्य) class). It is famous for **Chambal Ravines** (badlands), not the Damodar fault-valley tag.
- The **Betwa** rises in the Vindhya. The **Ken** joins near Banda.

- A west-to-east right-bank order stem that lists Betwa, Ken, Sind, and Chambal as items 1–4 answers with Chambal first, then Sind, then Betwa, then Ken.



> **Logic:** Ramganga at Kannauj is **true**. Betwa at Prayagraj is **false**. The **Padma** is the Ganga in Bangladesh. The **Jamuna** is the Brahmaputra in Bangladesh.

Panch Prayag and Uttarakhand



- Upstream to downstream on the Alaknanda the order is **Vishnuprayag** (विष्णुप्रयाग), then **Nandaprayag**, then **Karnaprayag** (कर्णप्रयाग), then **Rudraprayag** (रुद्रप्रयाग), then **Devprayag**.

Prayag	Confluence
Vishnuprayag	Alaknanda + **Dhauli Ganga** (western / Alaknanda Dhauli — not Eastern Dhauliganga of Kali)

Prayag	Confluence
Nandaprayag	Alaknanda + **Nandakini**
Karnaprayag	Alaknanda + **Pindar** (पिंडर)
Rudraprayag	Alaknanda + **Mandakini** (Kedarnath belt)
Devprayag	Alaknanda + **Bhagirathi** = **Ganga**

- **Eastern Dhauliganga** is a tributary of the **Kali / Sharda**, not of the Alaknanda Prayag set.
- **Mandakini** flows between Kedarnath and Rudraprayag.
- **Tehri** (टिहरी) is **Bhagirathi + Bhilangana**. It is a dam site, not one of the five Prayags.

PYQ — UPPCS Prelims 2024, Q74

Match List-I with List-II:

A. Devprayag · B. Rudraprayag · C. Karnaprayag · D. Tehri

1. Alaknanda and Pindar · 2. Alaknanda and Bhagirathi · 3. Bhagirathi and Bhilangna · 4. Alaknanda and Mandakini

A. 4 1 2 3

B. 2 3 1 4

C. 4 1 3 2

D. 2 4 1 3

Show answer

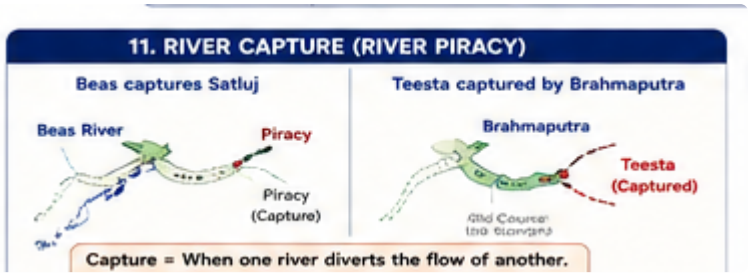
Ans: D — 2 4 1 3: Dev-2, Rudra-4, Karn-1, Tehri-3.

- The western **Ramganga** rises in the Doodhatoli hills and is regulated at Kalagarh.

Brahmaputra, capture, Barak

- The Brahmaputra rises as **Yarlung Tsangpo** near **Chemayungdung / Angsi** (Mansarovar region of Tibet). **Indus, Sutlej, and Brahmaputra** all have sources near Mansarovar / Kailash.
- It flows east in Tibet, then makes a sharp **U-turn** around **Namcha Barwa**. That bend is the eastern **syntaxial bend** of the Himalaya.
- It enters India west of Sadiya as **Siang / Dihang**. **Dibang** and **Lohit** join; thereafter it is the Brahmaputra in Assam.
- In Assam the channel is **braided**. **Majuli** is the world's largest inhabited river island (and India's first island district).
- Looking downstream in Assam (mostly westward flow): **right-bank (north)** tributaries include **Subansiri, Kameng, Manas, Sankosh, Teesta**. **Left-bank (south)** tributaries include **Burhi Dihing, Dhansiri, Kopili**.
- The **Teesta** rises in Sikkim (Khangchung Chho class — **not** the same source as the Brahmaputra). **Rangeet** is a Teesta tributary from Sikkim. Teesta was once a Ganga tributary; **river capture** shifted

it to the Brahmaputra.

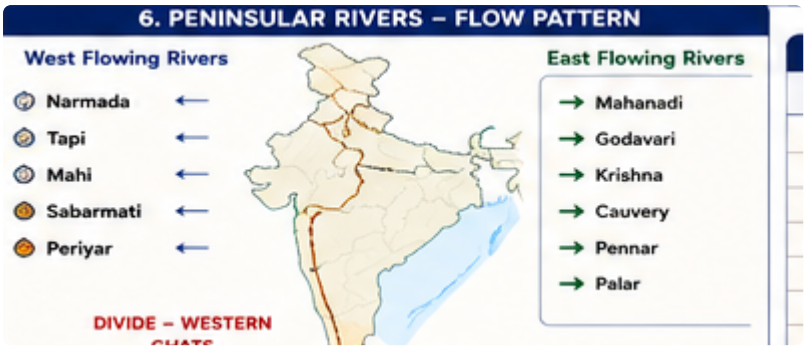


- Near Dhubri the river enters Bangladesh. There the **Tista** joins, and the river is called the **Jamuna** (not the Yamuna).
- The **Barak** rises in the Manipur hills (not Arunachal like Lohit/Subansiri). It becomes Surma–Kushiyara, then the **Meghna**, and reaches the **Bay of Bengal**. It is **not** inland drainage.
- Padma is joined by Jamuna, then by Meghna; the combined Sundarbans delta lies between the Hooghly and the Meghna. The Bengal Delta is subdivided into **moribund, mature, and active** belts.

> **Logic:** ** Padma = Ganga in Bangladesh. Jamuna = Brahmaputra in Bangladesh. Do not swap them with the Yamuna.

3.2 Peninsular rivers





- The main peninsular water divide is the **Western Ghats**. Most large rivers flow **east** to the Bay of Bengal and build **deltas**.
- **Narmada, Tapi, and Mahi** flow **west** to the Arabian Sea and mostly build **estuaries**, not deltas. They run short, steep courses through hard rock / rift valleys and carry less delta-building silt.
- Peninsular rivers are mostly rain-fed, with fixed courses and fewer meanders than Himalayan rivers. Narmada and Tapi in rift valleys are the main structural exceptions.

Eastern Ghats vs Chotanagpur Origins

- **Rivers originating from Eastern Ghats: Nagavali** and **Vamsadhara** rise directly from the Eastern Ghats.
- **Rivers originating from Chotanagpur Plateau: Subarnarekha** (rises near Nagri village in Ranchi district; Hundru falls; joined by Kharkai and Kanchi), **Damodar** (rises from Khamarpat hill), and **North/South Koel**.
- **Mahanadi Confluences & Classification:** The holy **Chandrahasini Devi Temple** is situated at the confluence of the **Mand** and **Mahanadi** rivers at Chandrapur (Chhattisgarh). In S.P. Chatterjee's 8-fold physiographic scheme of India, the Mahanadi Basin is grouped under the **Northern Deccan Plateau**.
- **Gandikota Canyon:** Known as the "Grand Canyon of India", sculpted through quartzite rocks by the **Pennar River** in Kadapa (YSR) district of Andhra Pradesh.
- **Kolleru Lake:** Freshwater lake in Andhra Pradesh situated between the Godavari and Krishna deltas; it is fed directly by the **Budameru** and **Tammileru** streams (indirectly connected to Krishna and Godavari systems).
- **Damodar Rift Valley & Coal Belts:** Flows through a downwarped fault valley across Jharkhand and West Bengal. Major tributaries include **Barakar**, **Jamunia**, **Barki**, and **Konar**. The **Eden Canal** (1938) was constructed from the Damodar for irrigation. The river valley hosts over 90% of India's Gondwana coal reserves (along with Son, Mahanadi, and Godavari valleys). Between Giridih and Durgapur, heavy industrial and mining effluents have turned the Damodar into a **biological desert**.

Godavari

- The Godavari is the longest peninsular river (about **1465 km**) and the largest peninsular basin. It is called **Dakshin Ganga** / **Vriddha Ganga**.
- It rises at **Trimbakeshwar, Nashik**, only about 80 km from the Arabian Sea, then flows east to the Bay.
- Left-bank tributaries are larger. **Penganga, Wardha, and Wainganga** meet as the **Pranhita**. **Indravati** and **Sabari** also join on the left.

- The **Manjra** is the main **right-bank** tributary (Nizam (निजाम) Sagar).
- **Pranhita belongs to the Godavari**, not the Mahanadi.
- Below Rajahmundry (राजमहेन्द्रवरम) the river splits into **Gautami** and **Vashishta** and builds a **lobate** delta.
- **Jayakwadi, Sriram Sagar, and Polavaram** are the projects.

Krishna

- The Krishna is the second-largest east-flowing peninsular river (about **1400 km**, basin about **2.59 lakh km²**).
- It rises just north of **Mahabaleshwar**.
- **Right bank:** Ghataprabha, **Malaprabha**, **Tungabhadra** (तुंगभद्रा).
- **Left bank:Bhima**, **Musi** (Hyderabad), Munneru.
- **Tungabhadra** is Tunga + Bhadra from the central Sahyadri.
- **Srisailam** and **Nagarjuna (नागार्जुन) Sagar** (नागार्जुन सागर) stand on the **Krishna**, not on the Tungabhadra.
- **Koyna** is a small tributary famous for its dam.
- The Krishna and Godavari deltas **merge** on the Andhra coast.

Mahanadi

- The Mahanadi rises at **Sihawa / Dandakaranya** in Chhattisgarh (about **851 km**, basin about **1.42 lakh km²**).
- Its upper course is the saucer **Chhattisgarh Plain**.
- **Left bank:** Seonath, Hasdeo, Mand, Ib.
- **Right bank:** Ong, **Tel**, **Jonk**.
- **Hirakud** (हीराकुंड) at Sambalpur is the great dam. **Cuttack** sits on the delta. The mouth is near **False Point**.
- In water and flood, Mahanadi ranks **second to Godavari** among peninsular rivers.

Kaveri (Cauvery)

- The Kaveri rises at **Talakaveri on Brahmagiri**, Coorg (about **765–800 km**). Some papers call it the **Ganga of the South**; Godavari remains **Dakshin / Vriddha Ganga**.
- **Hemavati, Kabini, Bhavani, and Amravati** are the usual tributaries.
- It forms the islands of **Srirangapatna, Shivasamudram, and Srirangam**. **Shivasamudram** is also the great Kaveri waterfall by volume.
- **Mettur (Stanley)** is on the Kaveri, **not** on the Krishna.
- The Kaveri delta is old and heavily irrigated. In **landform match-lists**, that ageing / abandoned-lobe story is why many keys tag **Cauvery** as a **moribund lobe**. Separately, UPPCS **"moribund subdivision"** stems still answer **Bengal Delta** — do not use Kaveri for that wording.
- Kaveri water is a **Karnataka–Tamil Nadu (नाडु)** dispute. Basin also touches Kerala and Puducherry.

Other east-flowing rivers

- The **Pennar** (पेन्नार) (Penner) drains Andhra–Karnataka to the Bay. It is **east-flowing**.

- The **Palar** and **Ponnaiyar** are Tamil Nadu east-flowing rivers.
- The **Vaigai** drains Madurai (मदुरै) to the Bay.
- The **Subarnarekha** rises on Chotanagpur (Ranchi). **Hundru Falls** is on it. It reaches the Bay.
- The **Brahmani–Baitarani** pair drains Odisha (NW-5 country).
- Of Periyar / Pennar / Palar, only **Periyar** reaches the Arabian Sea.
- Standard origin pairs: Yamuna–Yamunotri · Krishna–Mahabaleshwar · Godavari–Nashik · Mahanadi–Sihawa.

7. ORIGIN OF MAJOR PENINSULAR RIVERS

River	Origin
Yamuna	Yamunotri Glacier (Uttarakhand)
Ganga	Gangotri Glacier (Bhagirathi)
Godavari	Trimbakeshwar (Maharashtra)
Krishna	Mahabaleshwar (Maharashtra)
Mahanadi	Sihawa Hills (Chhattisgarh)
Narmada	Amarkantak (MP)
Tapi	Multai (Satpura, MP)
Kaveri	Brahmagiri Hills (Karnataka)
Luni	Pushkar (Aravalli, Rajasthan)

PYQ — UPPCS Prelims 2018, Q102

Which of the following rivers of India does NOT originate from the Western Ghats?

A. Godavari

B. Tapti/Tapi

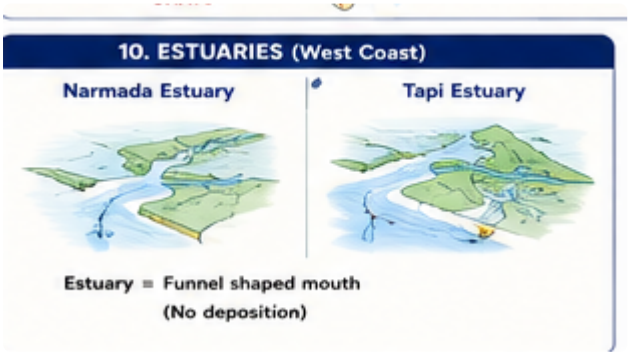
C. Kaveri

D. Kabini *(paper: Kabam in some OCR)*

Show answer

Ans: B — Tapi = **Multai (Satpura)**.

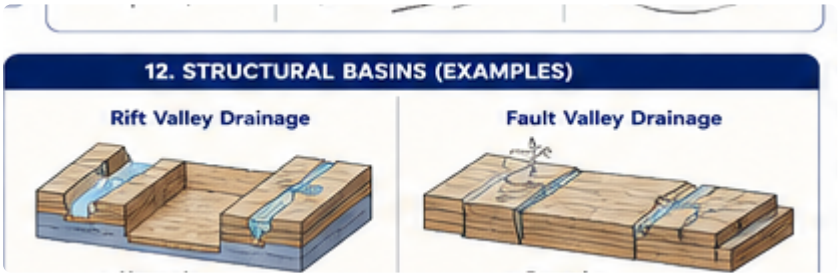
Narmada, Tapi, and other west-flowing rivers



- The **Narmada** is the **longest west-flowing** peninsular river (about **1312 km**). It rises at **Amarkantak** (Maikal) and uses the **rift** between Vindhya (north) and Satpura (south).
- It falls as **Dhuandhar** at the **Marble Rocks** near Jabalpur. It reaches the **Gulf of Khambhat**. **Aliabet** is the large estuary (ज्वारनदमुख) island.

- Right-bank tributaries are few. **Hiran** is the usual named right-bank stream. **Tawa** is a major left-bank tributary (joins near the Narmadapuram / Hoshangabad belt).
- Major dams on the Narmada, roughly upstream to downstream:
- **Bargi** is in Madhya Pradesh near Jabalpur. It is an upper Narmada multipurpose dam.
- **Indira Sagar** is in Madhya Pradesh (Khandwa / Punasa belt). It is among the largest reservoirs on the Narmada chain.
- **Omkareshwar** (ओंकारेश्वर) is also in Madhya Pradesh, downstream of Indira Sagar, near the Omkareshwar Jyotirlinga belt.
- **Sardar Sarovar** (सरदार सरोवर) is in Gujarat (Narmada district, near Navagam). It is the terminal multipurpose project of the Narmada Valley Development. Gujarat, Madhya Pradesh, Maharashtra, and Rajasthan share benefits. Do not confuse it with Indira Sagar (MP) or with Tapi's **Ukai**.
- The **Tapi (Tapti)** is the Narmada's "twin." It rises at **Multai** in the **Satpura** (सतपुड़ा), **not** on the Western Ghats. Length is about **724 km**. **Surat** sits near its mouth. **Ukai Dam** stands on the Tapi in Gujarat.
- The **Mahi** (माही) rises on the Vindhya in Dhar district. It crosses Madhya Pradesh, Rajasthan, and Gujarat to the Gulf of Khambhat.
- The **Sabarmati** rises in the Aravalli (अरावली) (Udaipur (उदयपुर) district). **Ahmedabad** (अहमदाबाद) and **Gandhinagar** sit on it.
- Hundreds of short **Sahyadri torrents** drop west. The **Sharavati** makes **Jog / Gersoppa Falls**.
- The **Periyar** is Kerala's great west-flowing river.
- **Mandovi** and **Zuari** are Goa (गोवा)'s estuary rivers.

Inland and fault-valley drainage



- The **Luni** ("salt river") rises on the western Aravalli near **Ajmer** (अजमेर). It dies in the **Rann of Kachchh**. Water is **fresh in the upper course** and **saline below Balotra**.
- The **Ghaggar** (घग्गर) is a seasonal Shiwalik (शिवालिक) stream. It fades in the Thar (थार) near Hanumangarh. It is the usual **Saraswati (सरस्वती) palaeo-channel**. It is **inland**.
- **Barak is not inland.** It reaches the Bay through the Meghna.
- **Fault / rift rivers:Narmada, Tapi, Damodar**. The phrase "famous for fault valley drainage" keys **Damodar**. Chambal is ravines, not that tag.

3.3 Rivers of Uttar Pradesh

Identity: Almost all of Uttar Pradesh sits in the **Ganga basin** (गंगा). The master streams are the **Ganga** and the **Yamuna**. Left-bank Himalayan feeders enter from the north and east. Right-bank peninsular

feeders enter the Yamuna from Bundelkhand and the Chambal belt.

Ganga in Uttar Pradesh

The **Ganga** is **formed in Uttarakhand** at Devprayag. It is **not** born inside Uttar Pradesh.

The river enters the UP plain in the **Bijnor** belt after Haridwar.

In Uttar Pradesh it passes **Kannauj, Kanpur (कानपुर), Prayagraj, Varanasi (वाराणसी), Ghazipur and Ballia**.

It leaves Uttar Pradesh into Bihar after the Ballia–Ghazipur fringe.

At **Prayagraj** the Ganga meets the Yamuna at the **Sangam** (संगम) (with the mythical Saraswati in tradition).

Yamuna in Uttar Pradesh

The **Yamuna** rises at **Yamunotri** on Bandarpunch in Uttarakhand.

It forms much of western Uttar Pradesh's border story after the Haryana–Delhi reach.

Mathura (मथुरा) and **Agra** (आगरा) stand on the Yamuna.

The Yamuna joins the Ganga at the Sangam in **Prayagraj**.

Left-bank feeders of the Ganga in / near UP

The **Ramganga** rises in the Doodhatoli / Garhwal hills of Uttarakhand.

It irrigates western Uttar Pradesh and meets the Ganga near **Kannauj**.

The Ramganga Project dam is about **825.8 m** long and **198 MW** in Garhwal.

The "**6 lakh ha** irrigated" figure is the **NOT matched** fact on that stem.

The **Gomati** (गोमती) (Gomti) rises at **Gomat Taal / Fulhar** in **Pilibhit**.

It is a classic plains-born tributary of the Ganga.

Lucknow stands on the Gomati.

The Gomati meets the Ganga near the **Saidpur / Ghazipur** belt.

The **Ghaghara** (Karnali in Nepal; **Saryu** at Ayodhya) rises near the Mapchachungo glacier class origin in Tibet / Nepal.

It enters the Terai (तराई) near **Bahraich** and waters Ayodhya–Gonda country.

Ayodhya stands on the Ghaghara / Saryu.

The Ghaghara meets the Ganga near **Chhapra / Revilganj (Saran)** in Bihar, not inside a UP city sangam.

The **Rapti** (राप्ती) rises in the Nepal Himalaya.

It enters eastern Uttar Pradesh and passes **Gorakhpur** (गोरखपुर).

The Rapti joins the **Ghaghara** inside Uttar Pradesh. It does **not** join the Ganga as a direct master feeder.

The **Gandak** (Narayani in Nepal) rises in the Nepal Himalaya from the Kali Gandak–Trishuli set.

It marks the eastern **Uttar Pradesh–Bihar** fringe (Kushinagar (कुशीनगर)–Deoria side) after entering India near Valmikinagar.

The Gandak meets the Ganga at **Sonpur near Patna**. The confluence itself is in Bihar.

Right-bank / southern rivers linked to UP

The **Son** rises at **Amarkantak** in Madhya Pradesh.

It cuts the Vindhyan / Sonbhadra (सोनभद्र) fringe of southern Uttar Pradesh.

The **Rihand** joins the Son in **Sonbhadra** (सोनभद्र). **Govind Ballabh Pant Sagar** sits on the Rihand.

The Son meets the Ganga near **Maner / Patna** in Bihar. Do not invent a Son–Ganga sangam inside UP.

The **Chambal** rises near Janapav in Madhya Pradesh.

It meets the Yamuna near **Etawah**.

The National Chambal Sanctuary is **Uttar Pradesh + Madhya Pradesh + Rajasthan**. It is **not** in Haryana.

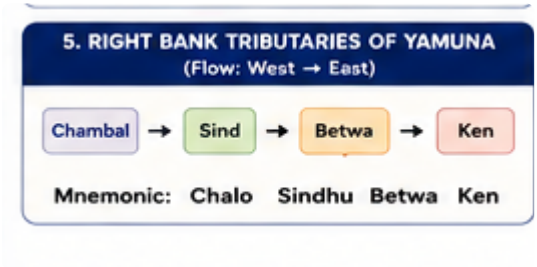
The **Betwa** rises in the Vindhya range of Madhya Pradesh.

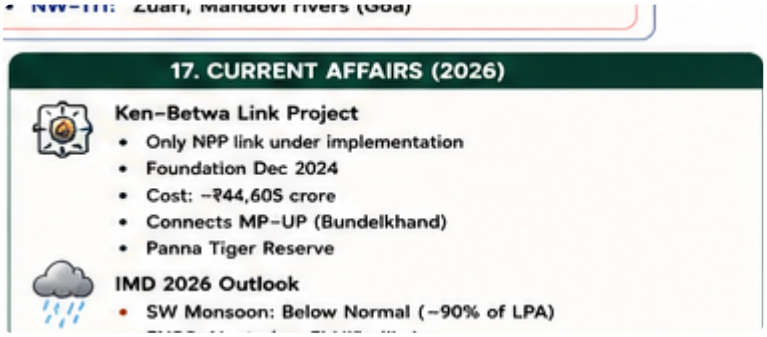
It meets the Yamuna at **Hamirpur**. It does **not** meet at Prayagraj.

The **Ken** rises in Madhya Pradesh and meets the Yamuna near **Banda**.

Ken–Betwa is the river-link under implementation for Bundelkhand water.

From west to east, the major Yamuna right-bank feeders in the UP story are **Chambal**, then **Sind**, then **Betwa**, then **Ken**.



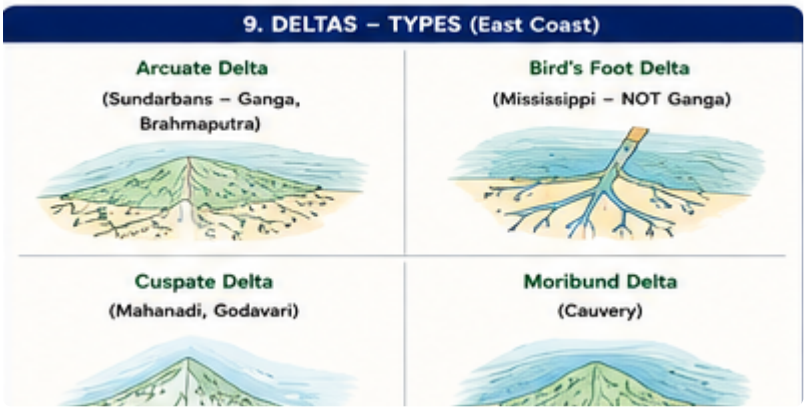


The **Hindon** is a Yamuna **left**-bank stream of western Uttar Pradesh / NCR.

3.4 River features and patterns

Deltas versus estuaries

Types of deltas — Teaching Card



- **Arcuate delta** has a convex seaward front and many distributaries spreading like a fan. The **Ganga–Brahmaputra (Sundarbans)** and **Godavari–Krishna** are Indian examples.
- **Bird's-foot (digitate) delta** has long finger-like distributaries. The **Mississippi** is the classic world example — **not** the Ganga.
- **Cuspate delta** forms when waves from two directions meet the river mouth. **Ebro** (Spain) is cited; **Mahanadi** is often used in Indian match-lists.
- **Lobate delta** is rounded with many shifting mouths. **Godavari** is sometimes classed here for its bulging front.
- **Moribund delta** means a dying / inactive part of a delta. Rivers there are silted, poorly connected to the main channel, and add little new land. Ox-bow lakes and marshes are common.
- Indian papers use **two frames**. Read the stem carefully.
- If the stem says **subdivision** of a named delta (UPPCS 2019 wording), the answer is the **Bengal (Ganga–Brahmaputra) Delta**, which is classed into **moribund, mature, and active** belts. The moribund belt lies mainly in the older western / north-western delta where distributaries decayed after main discharge shifted east toward the Padma.
- If the stem asks which river's delta is classed as a **moribund / abandoned lobe** in landform match-lists, coaching often keys the **Cauvery (Kaveri)** delta — an old, heavily irrigated delta with little active progradation.
- Do **not** merge the two into one memory line. Do **not** pick Krishna–Godavari or Mahanadi for either classic stem.
- **Estuary** replaces a delta where waves and tides dominate and the river mouth stays funnel-shaped — **Narmada, Tapi, Mandovi** on India's west coast.
- A **delta** grows where a river dumps silt faster than waves can remove it. The **east coast** has a wide shelf, so large deltas form.
- An **estuary** is a partly enclosed tidal mouth where river and sea mix. The **west coast** is steep and wave-strong, and rift rivers carry less silt, so **Narmada, Tapi, and Mandovi** form estuaries rather than deltas.

- Short, fast west-flowing Western Ghat rivers also fail to build large deltas for the same reason.
- World's largest estuary class example is often the **St Lawrence (लॉरेंस)**. India's largest estuary mouths are on the **Narmada / Tapi** side.

Type	Meaning	example
Arcuate	Fan-shaped front, many distributaries	Ganga–Brahmaputra (Sundarbans), Nile, Godavari–Krishna
Lobate	Rounded bulge, many mouths	Godavari (often)
Bird's-foot	Long finger-like distributaries	**Mississippi** — **not** Ganga
Cuspate	Pointed front from opposing waves	Ebro; **Mahanadi** often cited for India
Moribund (Bengal subdivision stem)	Dying belt inside a large delta's moribund–mature–active scheme	**Bengal Delta** (UPPCS 2019)
Moribund (lobe / river match stem)	Abandoned / ageing delta lobe	**Cauvery** in many landform keys
Estuarine mouth	Tidal funnel, little delta build-up	Narmada, Tapi

Origins, cities, dams (one home)

River	Origin
Bhagirathi	Gomukh / Gangotri
Alaknanda	Satopanth (above Badrinath)
Yamuna	Yamunotri (Bandarpunch)
Indus	Tibet, near Kailash / Mansarovar
Sutlej	Rakas Tal, Tibet (Trans-Himalayan; enters near Shipki La)
Brahmaputra	Tibet / Angsi–Chemayungdung (Tsangpo)
Godavari	Nashik / Trimbakeshwar
Krishna	Mahabaleshwar
Mahanadi	Sihawa, Chhattisgarh
Kaveri	Brahmagiri / Talakaveri, Coorg
Narmada	Amarkantak
Tapi	Multai, Satpura
Son	Amarkantak
Chambal	Mhow / Ambedkar Nagar (Malwa)
Luni	Aravalli near Ajmer

River	Origin
Sharavati	Western Ghats (Jog Falls)
Subarnarekha	Chotanagpur (Ranchi / Nagri class)
Jhelum	Verinag (Pir Panjal)
City	River
------	-------
Lucknow	Gomti
Ayodhya	Ghaghara / Saryu
Prayagraj	Ganga + Yamuna (Sangam)
Kanpur	Ganga
Varanasi	Ganga
Patna	Ganga
Bhagalpur	Ganga
Surat	Tapi
Hyderabad	Musi
Vijayawada	Krishna
Ahmedabad	Sabarmati
Cuttack	Mahanadi
Bhubaneswar	Not on Mahanadi bank
Ujjain (उज्जैन)	Kshipra
Jabalpur	Narmada
Nasik (नासिक)	Godavari
Nanded	Godavari
Rajahmundry	Godavari
Srinagar	Jhelum
Leh (लेह)	Indus (right bank)
Ludhiana	Old Sutlej belt
Firozpur	Old Sutlej belt
Jamshedpur	Subarnarekha
Guwahati	Brahmaputra
Dibrugarh	Brahmaputra

River	Origin
Kota	Chambal
Gorakhpur	Rapti
Dam / project	River
---	---
Bhakra–Nangal	Sutlej
Pandoh	Beas (**not** Ravi)
Baglihar / Salal	Chenab
Ranjit Sagar	Ravi
Tehri	Bhagirathi
Farakka Barrage	Ganga (Hooghly diversion)
Hirakud	Mahanadi
Nagarjuna Sagar / Srisailam	Krishna (**not** Tungabhadra)
Mettur (Stanley)	Kaveri (**not** Krishna)
Jayakwadi	Godavari (Maharashtra)
Sriram Sagar	Godavari (Telangana)
Polavaram	Godavari (Andhra Pradesh)
Bargi	Narmada (MP, near Jabalpur)
Indira Sagar	Narmada (MP, Khandwa / Punasa)
Omkareshwar	Narmada (MP)
Sardar Sarovar	Narmada (Gujarat, Navagam) — terminal Narmada project
Ukai	Tapi (Gujarat)
Gandhi (गांधी) Sagar	Chambal
Rihand	Rihand (Son system)
Rajghat	Betwa (UP–MP)

> **Teaching note: Pandoh** stands on the **Beas**, not the Ravi. **Srisailam** stands on the **Krishna**, not the Tungabhadra. **Mettur (Stanley)** stands on the **Kaveri** (कावेरी), not the Krishna. Hyderabad is not a Krishna-stem dam site. **Sardar Sarovar** is on the **Narmada** in Gujarat; it is not Indira Sagar in Madhya Pradesh and not Ukai on the Tapi.

Additional City / Town	River	Key Geographic Anchor
Pahalgam	**Lidder**	Starting point of Amarnath Yatra in Kashmir (2,130 m altitude)
Pandharpur	**Chandrabhaga (Bhima)**	Famous pilgrimage centre of Vithoba in Maharashtra
Hampi	**Tungabhadra**	Historic capital of Vijayanagara Empire (not Malaprabha)
Govindghat	**Alaknanda + Lakshman Ganga**	Confluence point en route to Hemkund Sahib and Valley of Flowers
Gaurikund / Guptakashi / Rambara	**Mandakini**	Pilgrim transit halts along the Mandakini gorge to Kedarnath
Bhubaneswar	**Kuakhai** (delta branch)	Southwest of Mahanadi main stem; Cuttack is on Mahanadi
Kanpur	**Ganga**	Largest urban agglomeration (>2.9 million) on the bank of Ganga
Leh	**Indus (Right Bank)**	Situated strictly on the right bank of the Indus in Ladakh

Drainage patterns, rejuvenation, basins — Teaching Card

A **drainage pattern** is the map shape of rivers. The shape tells you what **controls** the ground — rock type, folds, joints, or slope.

Tree and grid patterns

- **Dendritic** looks like a **tree** with many irregular branches.
- It develops on **homogeneous / uniform rock** — the same resistance everywhere, so streams join at acute angles without a forced grid.
- The **Ganga (गंगा)** on the northern plain follows a dendritic pattern.
- The **Indus (सिंधु)** on the plain follows a dendritic pattern.
- The **Godavari (गोदावरी)** follows a dendritic pattern.
- The **Mahanadi (महानदी)** follows a dendritic pattern.
- The **Krishna (कृष्णा)** follows a dendritic pattern.
- The **Kaveri (कावेरी)** follows a dendritic pattern.
- **Trellis** looks like a **garden trellis / lattice** — long parallel master streams with short cross-links.
- It develops on **folded** belts of **hard and soft rock** alternating — soft belts become strike valleys; hard belts become ridges.
- The Indian trellis ground is the old folded belt of **Singhbhum (सिंहभूम)** on the **Chotanagpur (छोटानागपुर)** plateau.
- Memory: **same rock → tree (dendritic)**; **folds → lattice (trellis)**.

From a high centre or into a bowl

- **Radial** streams run **outward** from a high centre — like spokes from a hub.
- Typical Indian controls are a dome, a volcano, or a highland such as **Amarkantak** or **Girnar**.
- The **Narmada (नर्मदा)** leaves Amarkantak toward the **west**.
- The **Son (सोन)** leaves Amarkantak toward the **Ganga**.
- The Mahanadi rises at **Sihawa**, so it is not an Amarkantak radial river.
- **Centripetal** is the opposite — streams run **inward** into a **closed basin**.
- Streams of the **Imphal** basin drain inward toward **Loktak (लोकतक)**.
- The **Mendha (मेंढा)** drains inward toward **Sambhar (सांभर)** lake.
- The **Rupangarh (रुपनगढ़)** drains inward toward Sambhar lake.

Joints and steep slopes

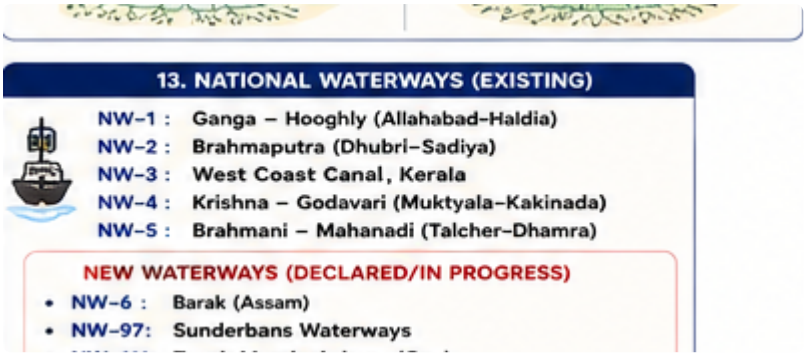
- **Rectangular** drainage bends at near-right angles because streams follow **joints** and **faults**.
- Streams on the jointed **Vindhyan (विंध्य)** rocks follow a rectangular pattern.
- **Parallel** drainage has many streams running side by side on a **steep, uniform slope**.
- The **Sharavati (शरावती)** follows a parallel pattern down the Western Ghats.

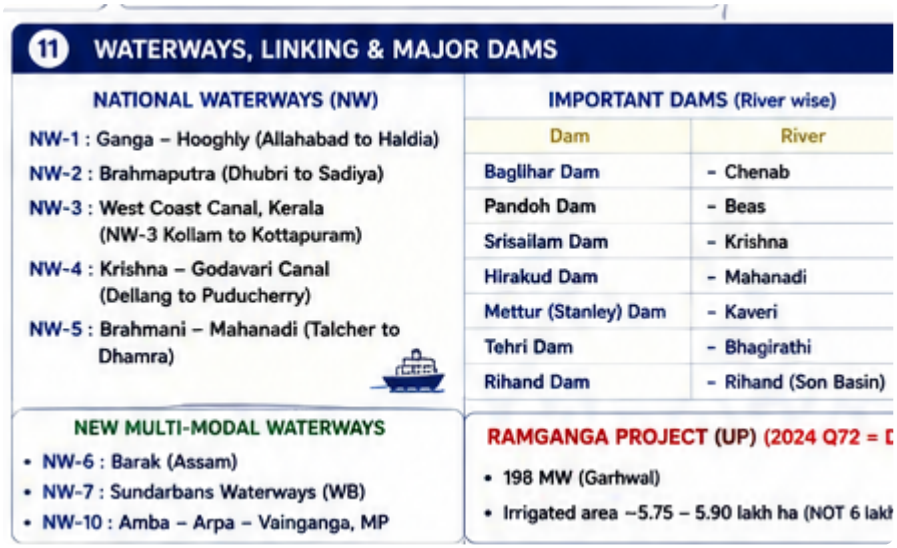
Pattern	Control	River / ground
Dendritic	Uniform rock	**Ganga** and **Indus** plains; **Godavari**; **Mahanadi**; **Krishna**; **Kaveri**
Trellis	**Folded** ridges and valleys	**Singhbhum** (Chotanagpur)
Radial	Dome or peak	**Narmada** west and **Son** toward the Ganga from **Amarkantak**; Girnar
Centripetal	Closed basin	**Loktak** (Imphal); **Mendha** and **Rupangarh** into **Sambhar**
Parallel	Steep uniform slope	**Sharavati** (Western Ghats)
Rectangular	Joints and faults	Jointed **Vindhyan** streams

- Youth shows gorges and waterfalls. Maturity shows meanders and floodplains. Old age shows ox-bows and deltas. The Brahmaputra is **braided**.
- **Rejuvenation** is renewed downcutting after uplift, base-level fall, or capture. Knickpoints, terraces, and incised meanders appear.
- A **structural basin** is an enclosed lowland, often centripetal (Kashmir vale, Imphal).

> **Teaching note:** Folded structures (जाली) produce a **trellis** pattern. They do not produce dendritic, radial, or rectangular patterns as the first answer.

3.5 Waterways and river linking



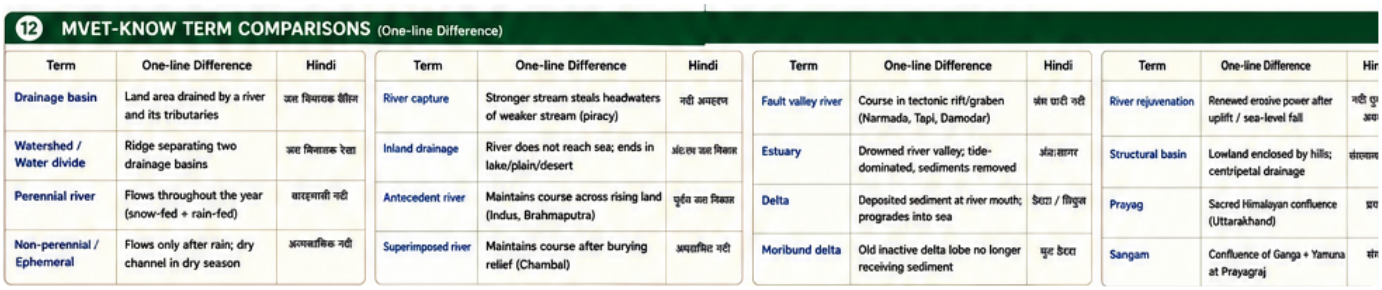


NW	Route
NW-1	Ganga–Bhagirathi–Hooghly (Haldia–Prayagraj) — longest
NW-2	Brahmaputra (Dhubri–Sadiya)
NW-3	Kerala west-coast canal / backwaters
NW-4	Krishna–Godavari canals
NW-5	Brahmani–Mahanadi–Paradip

- The National Waterways Act **2016** declared **106** waterways. Core remains NW-1 to NW-5. Other named waterways: **NW-16 Barak**, **NW-97 Yamuna**, **NW-37 Gandak**.
- The most navigable rivers are the **Ganga, Brahmaputra, and Mahanadi**. Godavari, Krishna, Narmada, and Tapi are navigable mainly near their mouths.
- **Interlinking (NPP):** NWDA / Jal Shakti. The plan lists **16 peninsular + 14 Himalayan** links. The aim is surplus to deficit, plus flood, drought (सूखा), power, and navigation.
- **Ken–Betwa** is the **only priority NPP link under implementation** (Cabinet 2021; foundation **25 December 2024**). Both rivers are in the **Yamuna** basin. The project is **Madhya Pradesh–Uttar Pradesh Bundelkhand**. **Daudhan dam** and **Panna Tiger Reserve** submergence are the ecology facts.
- Polavaram includes Godavari–Krishna link assistance.

- Disputes: **Cauvery** (Karnataka–Tamil Nadu). **Krishna** (Maharashtra–Karnataka–Telangana–Andhra Pradesh). **Ravi–Beas / SYL** (Punjab–Haryana). **Narmada** (Gujarat–Madhya Pradesh–Maharashtra–Rajasthan). Law: ISRWD Act 1956 (अधिनियम) and Article **262**.
- Namami Gange and GAP (1985) clean the Ganga on the NW-1 stretch.

📝 Master Drainage Comparisons & Memory Hooks



12 MVET-KNOW TERM COMPARISONS (One-line Difference)

Term	One-line Difference	Hindi
Drainage basin	Land area drained by a river and its tributaries	जल विभाजक क्षेत्र
Watershed / Water divide	Ridge separating two drainage basins	जल विभाजक रेखा
Perennial river	Flows throughout the year (snow-fed + rain-fed)	बारहमासी नदी
Non-perennial / Ephemeral	Flows only after rain; dry channel in dry season	अल्पकालिक नदी

Term	One-line Difference	Hindi
River capture	Stronger stream steals headwaters of weaker stream (piracy)	नदी अपहरण
Inland drainage	River does not reach sea; ends in lake/plain/desert	अंतःस्थ जल निकास
Antecedent river	Maintains course across rising land (Indus, Brahmaputra)	पूर्वत जल निकास
Superimposed river	Maintains course after burying relief (Chambal)	अध्यारोपित नदी

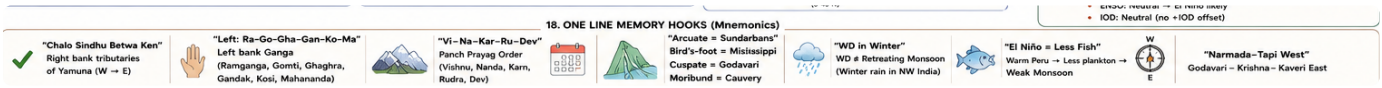
Term	One-line Difference	Hindi
Fault valley river	Course in tectonic rift/graben (Narmada, Tapi, Damodar)	भ्रंश घाटी नदी
Estuary	Drowned river valley; tide-dominated, sediments removed	ज्वारनदमुख
Delta	Deposited sediment at river mouth; progrades into sea	डेल्टा / त्रिभुज
Moribund delta	Old inactive delta lobe no longer receiving sediment	मृत डेल्टा

Term	One-line Difference	Hir
River rejuvenation	Renewed erosive power after uplift / sea-level fall	नदी पु अप
Structural basin	Lowland enclosed by hills; centripetal drainage	संरचना
Prayag	Sacred Himalayan confluence (Uttarakhand)	प्र
Sangam	Confluence of Ganga + Yamuna at Prayagraj	सं



RAPID FACTS (RAATA)

- Largest basin in India – Ganga
- Largest peninsular basin – Godavari
- Longest peninsular river – Godavari (~1465 k
- Longest west flowing – Narmada (~1312 km)
- Only river in Kerala flowing to Arabian Sea – Periyar
- Only trellis pattern – in folded mountains
- Radial pattern centre – Amarkantak
- Only moribund delta – Cauvery
- Teesta – captured by Brahmaputra
- Jonk – Mahanadi tributary, NOT Ganga
- Sutlej – Trans-Himalayan (MCQ trap)
- Rann of Kutch – formed by Luni



18. ONE LINE MEMORY HOOKS (Mnemonics)

"Chalo Sindhu Betwa Ken" — Right bank tributaries of Yamuna (W → E)

"Left: Ra–Go–Gha–Gan–Ko–Ma" — Left bank Ganga (Ramganga, Gomti, Ghaghra, Gandak, Kosi, Mahananda)

"Vi–Na–Kar–Ru–Dev" — Panch Prayag Order (Vishnu, Nanda, Karn, Rudra, Dev)

"Arcuate = Sundarbans" / Bird's-foot = Mississippi / Cuspate = Godavari / Moribund = Cauvery

"WD in Winter" — WD ≠ Retreating Monsoon (Winter rain in NW India)

"El Niño = Less Fish" — Warm Peru → Less plankton → Weak Monsoon

"Narmada–Tapi West" — Godavari – Krishna – Kaveri East

IOD: Neutral (no +IOD offset)
